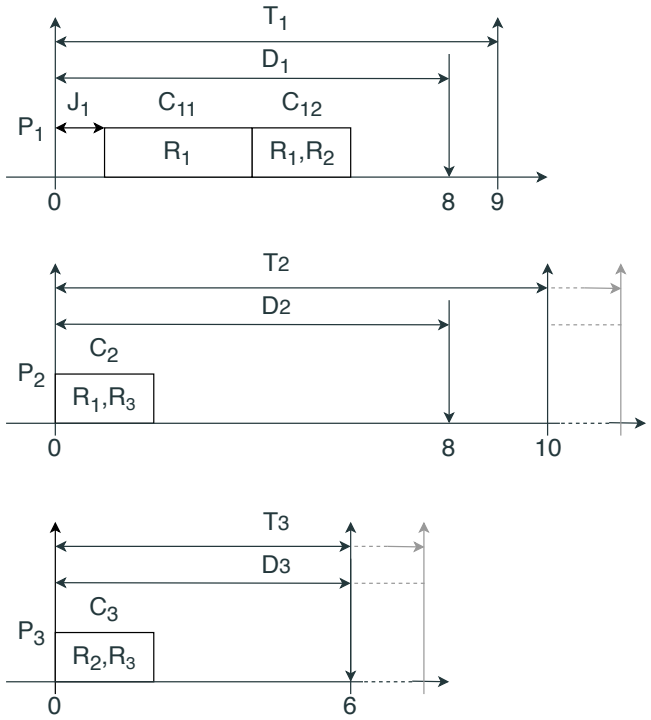
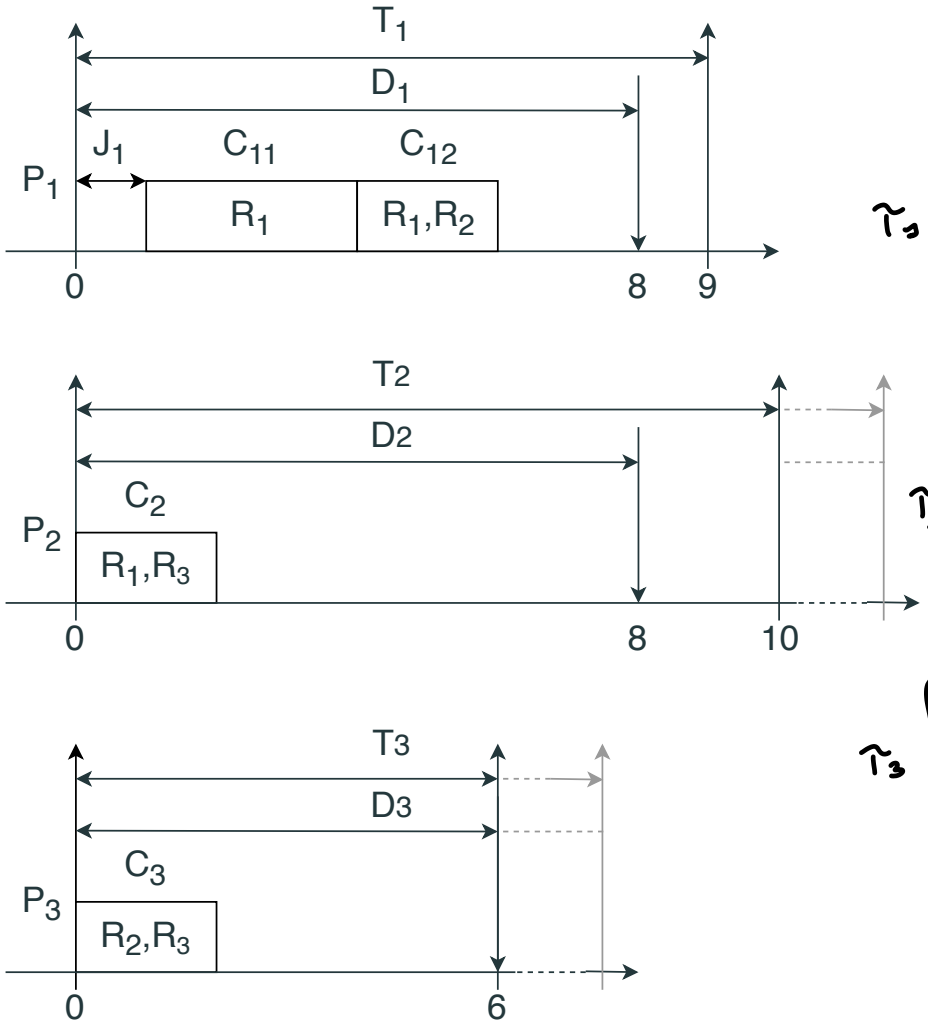
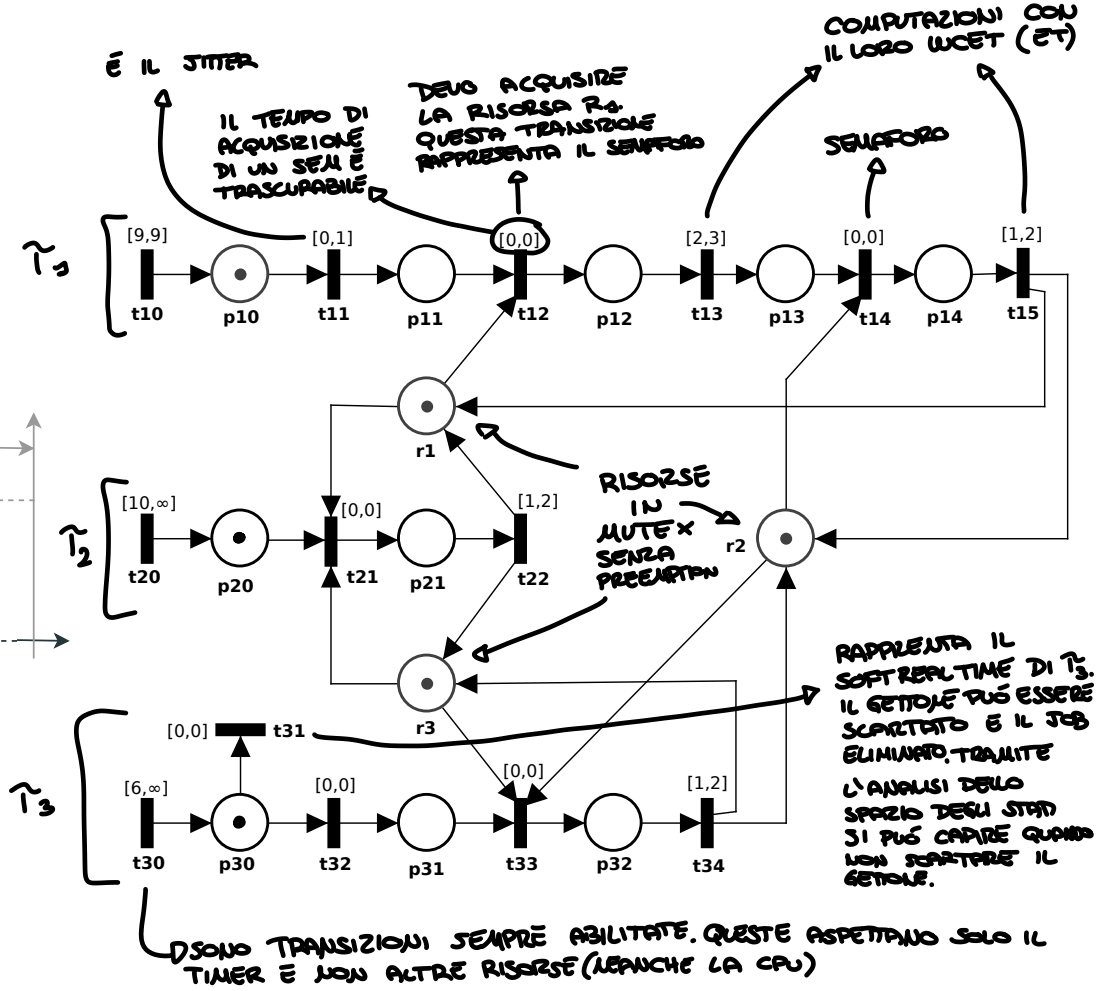
Modeling real-time task-sets with TPNs: an example (1/4)

Quando un task acquisisce una risorsa non può subire preemption, cioè non rilascia la risorsa finché non ha finito il suo compito.

- 3 tasks P_1, P_2, P_3 contending for 3 **non-preemptable** resources R_1, R_2, R_3
  - P_1 and P_2 are hard real-time (mandatory), P_3 is soft real-time (optional)
- P_1 is periodic
  - period $T_1 = 9$
  - relative deadline $D_1 = 8$
  - absolute start time jitter $J_1 \leq 1$
  - computation 1: WCET $C_{11} \in [2, 3]$, uses R_1
  - computation 2: WCET $C_{12} \in [1, 2]$, uses R_1, R_2

    SAREBBE ET PERCHÉ IL TEMPO DI ESEC È IN [1,2]. WCET SARAREBBE 2.
- P_2 is sporadic
  - inter-arrival time $T_2 \geq 8$
  - relative deadline $D_2 = 8$
  - computation: WCET $C_{21} \in [1, 2]$, uses R_1, R_3
- P_3 is sporadic
  - inter-arrival time $T_3 \geq 6$
  - relative deadline $D_3 = 6$
  - computation: WCET $C_{31} \in [1, 2]$, uses R_2, R_3



Modeling real-time task-sets with TPNs: an example (2/4)

- The TPN could be automatically derived from the timeline